

Revisiting Group Contribution Theory for Estimating Fractional Free Volume of Microporous Polymer Membranes – Supplementary Information

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1. Example V_W calculations and discussion of usage of similar groups in Tables 1 to 3

The PIM-1 repeat unit can be broken into groups as such:

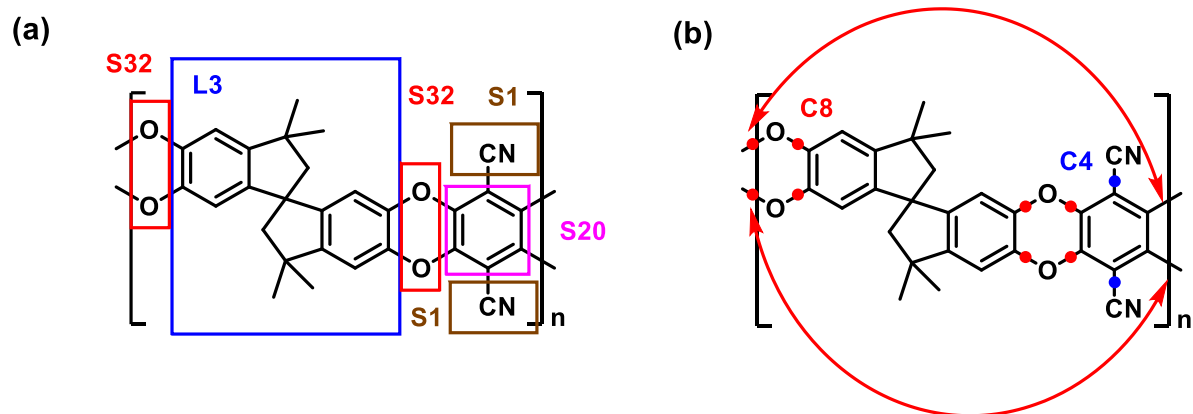


Figure S1: (a) Segmenting PIM-1 into its various groups. **Note that the largest group possible should be used for best results.** (b) The overlap volumes that should be subtracted in order to connect the groups. Each dot represents a connection between groups, of which the corresponding overlap volume should be subtracted. In this example, the red dots represent a $C_{ar}-O$ connection (C8) and the blue dots represent a $C_{ar}-C$ connection (C4). Care should be taken not to double count repeat unit end groups; the red arrows indicate equivalent connections that should not be double-counted.

To calculate V_W from these groups, the V_W for each group should be added together and the overlap volumes corresponding to the connections between the groups should be subtracted according to Equation 6 in the main text:

$$V_{W,tot} = \sum_{i=1}^n V_{W,i} - \sum_{j=1}^m V_{overlap,j}$$

Where n is the number of groups, m is the number of connections between groups, $V_{W,i}$ is the van der Waals volume for the groups from Tables 1–3, and $V_{overlap,j}$ is the overlap volume from Table 4 between groups. Following the example shown in Figure S1, V_W can be calculated as:

$$V_W = [2 \times S32 + L3 + 2 \times S1 + S20] - [8 \times C8 + 2 \times C4]$$

$$V_W = [2 \times 17.00 + 157.7 + 2 \times 16.57 + 43.10] - [8 \times 4.091 + 2 \times 4.688] = 225.836 \frac{cm^3}{mol}$$

For a typical PIM-1 with a density of 1.05 g cm^{-3} [6], the resulting FFV is:

$$FFV = 1 - 1.3 \frac{\rho}{MW} V_W = 1 - 1.3 \frac{1.05 \frac{g}{cm^3}}{460.63 \frac{g}{mol}} 225.833 \frac{cm^3}{mol} = 0.331$$

Another example calculation for KAUST-PI-1 is shown below

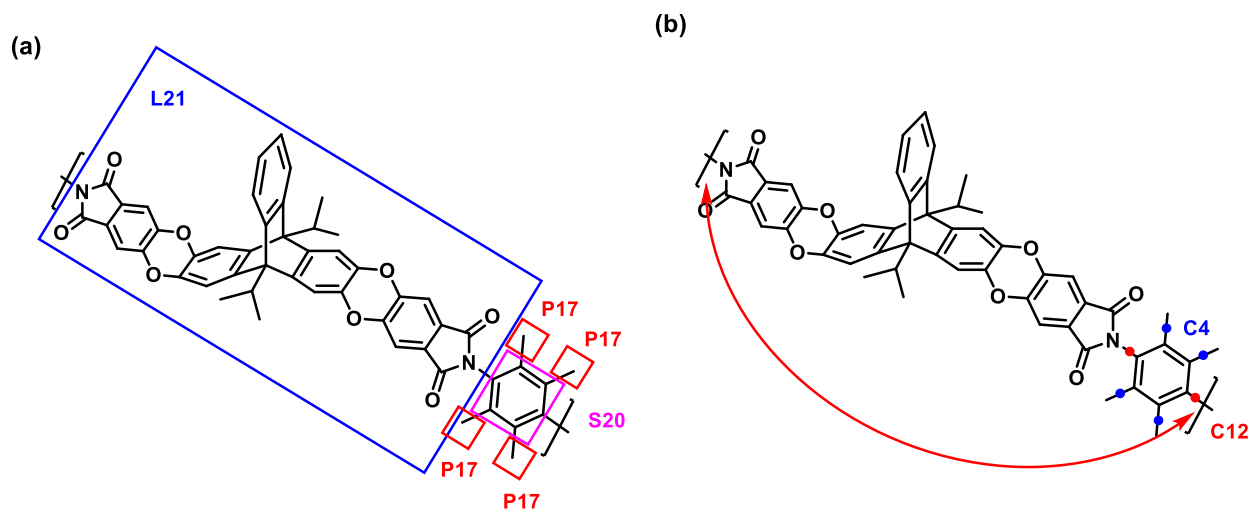


Figure S2: (a) Segmenting KAUST-PI-1 into its various groups. **Note that the largest group possible should be used for best results.** (b) The overlap volumes that should be subtracted in order to connect the groups. Each dot represents a connection between groups, of which the corresponding overlap volume should be subtracted. In this example, the red dots represent a C_{ar}-N connection (C12) and the blue dots represent a C-C_{ar} connection (C4). Care should be taken not to double count repeat unit end groups; the red arrow indicates equivalent connections that should not be double-counted

Again using Equation 6 in the main text, the V_W can then be calculated as:

$$V_W = [L21 + 4 \times P17 + S20] - [4 \times C4 + 2 \times C12]$$

$$V_W = [331 + 4 \times 14.63 + 43.10] - [4 \times 4.688 + 2 \times 4.255] = 405.358 \frac{cm^3}{mol}$$

For a density of 1.09 g cm^{-3} [17], the resulting FFV is:

$$FFV = 1 - 1.3 \frac{\rho}{MW} V_w = 1 - 1.3 \frac{1.09 \frac{g}{cm^3}}{818.95 \frac{g}{mol}} 405.358 \frac{cm^3}{mol} = 0.299$$

The reader may have noted that certain similar groups are present in Tables 2 and 3. For example, triptycene units are represented in entries S4 and L11, spirobisindane in entries S2 and L3, etc. The groups in Table 3 were tabulated for ease-of-use, as they represent the most common forms of the corresponding base unit in Table 2. However, there are some structures that do not fully conform with the most common form of the structural unit, especially as new structures are synthesized and characterized. A literature example is for vinylated PIM-1 as synthesized by Halder *et al.* [32] The segmented structure is shown below:

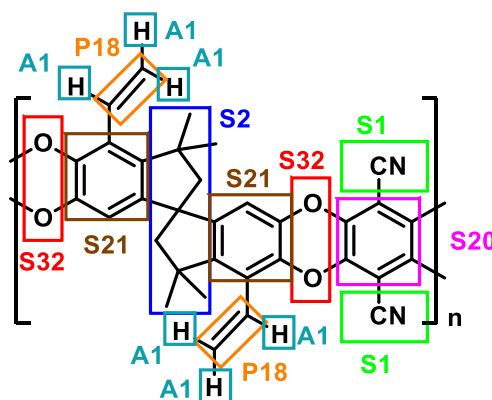


Figure S3: Segmenting vinylated PIM-1 identified with its various groups.

In this case, use of group L3 to build to spirobisindane group would not be appropriate because group L3 explicitly contains only four connections to other groups on the aromatic rings, while the structure for vinylated PIM-1 requires six total. In this case, use of the base unit S2 along with two S21 groups would be appropriate to build the spirobisindane group for this structure. An alternative approach to build this unit is to start with group L3, subtract the V_W of two hydrogen atoms (A1), then add the overlap volume corresponding to $C_{ar}-H$ (C2). While both approaches are valid, the method shown in Figure S3 is recommended to minimize the possibility of user error.

2. Example calculation comparing reporting methods for perfluoropolymers

As shown in Figure 1, the only three polymers that showed a decrease in calculated FFV when comparing the original Bondi method to our updated method were perfluoropolymers. Since these polymers do not contain any hydrogen atoms, the change in the hydrogen VDW radius does not have an effect for these polymers. Assumption 4 states “For unspecified atom-atom connections (i.e., connections between groups), V_W can be calculated by first assuming the group is connected to an aliphatic carbon atom and then subtracting the sphere cap of the connecting atom.” When considering bonds with heteroatoms (*e.g.*, C–O or C–F), this assumption results in a mismatch at the sphere cross-sectional interface as illustrated below. The mismatch at the interface is caused by the difference in bond length between a C–C bond (1.53 Å) and a C–F bond (1.36 Å), and therefore this issue will exist for connections to any bonds with heteroatoms.

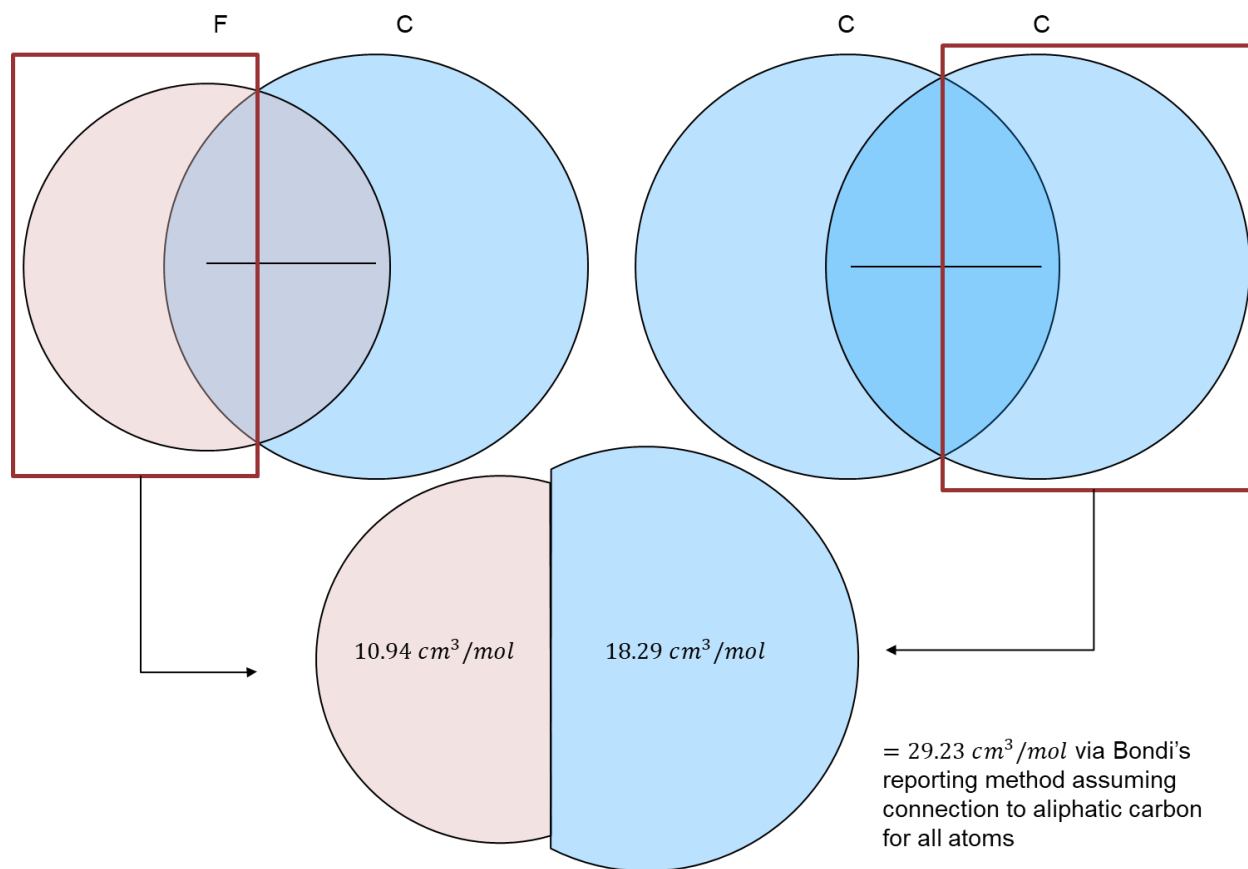
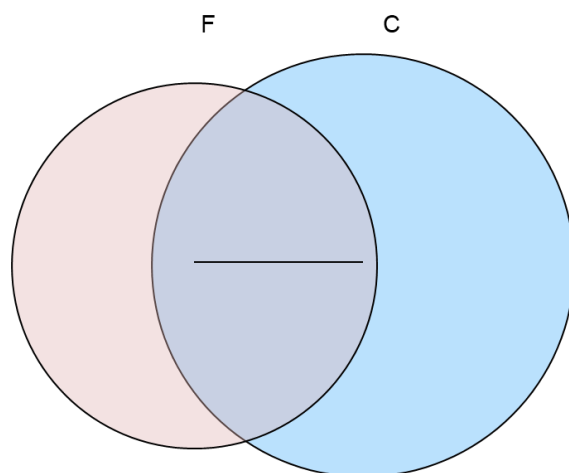


Figure S4: Schematic showing the effect of assumption 4 when considering bonds to heteroatoms. The circles representing the VDW volumes as well as the lines representing center-to-center bond lengths are drawn to scale. The center-to-center “bond length” of the resulting combination using Bondi’s reporting method is 1.19 Å, considerably shorter than either of the original C–C (1.53 Å) and a C–F (1.36 Å) bond lengths.

On the other hand, the reporting method proposed in this study does not assume any connection and therefore does not prematurely subtract a sphere cap volume, illustrated below.



New method:

$$V_w = V_C + V_F - V_{CF} = 29.98 \text{ cm}^3 / \text{mol}$$

Figure S5: Schematic showing the proposed reporting method, where the subtracted overlap volume was determined as a function of the two atoms and the bond length.

As shown in Figures S3 and S4, the new method results in a larger V_w compared to the old method by $0.75 \text{ cm}^3 \text{ mol}^{-1}$ per C–F bond. When propagated across the 10–12 bonds for Teflon AF 2400, Teflon AF 1600, and Hyflon AD 60 perfluoropolymers, the additional $7.5\text{--}9 \text{ cm}^3 \text{ mol}^{-1}$ resulted in a $\sim 14\%$ increase in the total V_w , which in turn lowered the calculated FFV by $\sim 17\%$.

3. Database of literature values

Table S1: Compiled database of polymers used in this study. Only samples with reported densities were considered. In some cases, the FFV or permeability of certain gases were not reported by the reference. Reported FFV with a “*” designation represents FFV obtained via MD simulation. Polymer names are shown as they appear in their corresponding reference.

Polymer Name	Density (g cm ⁻³)	Reported FFV	New FFV	P_{He} (barrer)	P_{H_2} (barrer)	P_{O_2} (barrer)	P_{CH_4} (barrer)	P_{N_2} (barrer)	P_{CO_2} (barrer)	Ref.
PTMSP	0.78		0.362	6200	16500	8800	17000	6400	35200	[1]
SBF-PIM	1.056		0.355	2200	6320	2640	1100	786	13900	[2]
PIM-Trip-TB	1.14		0.261	2500	8039	2718	905	629	9709	[3]
PIM-1	0.92		0.414					113.9	3027.7	[4]
PIM-1	1.18		0.248		2254		361.68	222	4521	[5]
PIM-1	1.05		0.331					283	4011	[6]
PIM-1	1.05		0.331		1681		497	275		[7]
PMDA-DAT	1.252	0.196	0.233		71.1	9.4	1.6	1.5	51.4	[8]
6FDA-DATRI	1.297	0.226	0.251	198	257	39	6.2	8.1	189	[9]
PIM-6FDA-OH	1.3		0.230		181	23.8	3.4	5.5	119	[10]
Co-80/20	1.312		0.235		285	32.6	4.1	6.8	173	[10]
Co-50/50	1.362		0.226		171	16.3	1.45	2.85	77	[10]
PIM-6FDA-OH	1.245		0.262		228	33.6	5.3	7.15	168	[10]
Co-80/20	1.25		0.271		380	49.1	6.7	10.2	261	[10]
Co-50/50	1.329		0.245		155	14.5	1.3	2.5	67	[10]
PIM-1	1.08	0.25	0.312	1500	3600	1300	430	340	6500	[11]
PIM-1	1.1	0.17	0.299			728	273	192	3815	[12]
PIM-1	1.09	0.238	0.305	947.4		585.5	317.7	168.1	3488.7	[13]
cPIM-1	1.2	0.182	0.262		1619	462.2	208.6	142.9	2654	[14]
PIM-1	1.112	0.23	0.291				310	228	3799	[15]
PMDA-TMID	1.158	0.206*	0.260	535		232	76	58	1190	[16]
KAUST-PI-1	1.09		0.299		4183	827		169		[17]

PIM-PI-1	1.15		0.266		530	150	77	47	1100	[17]
PIM-1-00	1.15		0.267			278.2	91.2	72.9	1104.7	[18]
TPI-PBO-0.25	1.409	0.204	0.225		878	197	34	53	952	[19]
TPI-PBO-0.50	1.381	0.2	0.228		762	141	25	34	667	[19]
TPI-PBO-0.75	1.361	0.193	0.227		574	107	18	25	505	[19]
PIM-1	1.1		0.299			576		178	3657	[20]
DPt-TMPD	1.1	0.201	0.259	490	1007	348	205	111	2035	[21]
DPt-MBDAM	1.11	0.183	0.242	285	542	158	86	52	932	[21]
DPt-BAPFH	1.23	0.168	0.225	177	237	56	21	16	320	[21]
(PIM-PI)x-b-(PI)y (x:y= 1:4)	1.12	0.2868*	0.342				184	176	3011	[22]
(PIM-PI)x-b-(PI)y (x:y= 1:6)	1.18	0.2585*	0.310				104	94	1660	[22]
(PIM-PI)x-b-(PI)y (x:y= 1:8)	1.23	0.2445*	0.282				63	80	1435	[22]
PIM-Trip-TB	1.02		0.339				310.501		4109.194	[23]
PIM-1	0.92		0.414					195.4	3694.5	[24]
TRIP-TR-460-30 (TR-6FDA-DAT1-OH)	1.15		0.329	478	791	170	40.1	50	840	[25]
6FDA-DAT1-OH	1.24		0.296	90	90	9.36	0.83	1.54	43	[25]
Bio-TBPI-1	1.093	0.21	0.272	314	669	127	22	27	575	[26]
Bio-TBPI-2	1.135	0.212	0.260	355	744	165	29	37	702	[26]
TDAl3-HB	1.07	0.293*	0.297	449	982	188	50	46.1	998	[27]
6FDA-HB	1.22	0.259*	0.291	271	391	62	10.9	14.2	286	[27]
6FDA-TrMCA	1.31	0.218*	0.260		193	26.4	3.2	5.52	144	[28]
CTB1-DMN	1.18		0.240		1295	320	95.7	76.2	1661	[29]
CTB2-DMN	1.2		0.248		1150	206	40.4	39.9	948	[29]
SBI-HTB	1.26		0.202		467	75.7	16.3	16.6	466	[30]
6FDA-HTB	1.46		0.182		167	13.6	0.92	2.26	67	[30]
6FDA-HTB	1.46		0.182		150	11.6	0.72	1.8	55	[30]
EAD-DMN	1.17		0.255		2856	655	255	171	3500	[31]
EA-DMN	1.18		0.233		3291	863	154	171	3321	[31]
PIM1-CO-100	1.05	0.25	0.319		3710	2180	1525	850	14180	[32]
PIM1-CO-50	1.07	0.249	0.312		2760	1630	1050	600	10990	[32]
vinylated PIM-1	1.12	0.187	0.268		1081	410	260	130	3240	[32]
vinylated PIM-1 CO50	1.15	0.186	0.257		1140	315	185	100	2480	[32]
thiophenated vinylated PIM-1 CO50	1.18	0.156	0.238		582	180	95	50	1370	[32]
PIM1-COBr-50	1.19	0.252	0.301		1790	710	370	220	5030	[32]
thiophenated vinylated PIM-1	1.21	0.151	0.209		775	234	125	70	1735	[32]

PIM1-COBr-100	1.29	0.243	0.307		1300	480	185	115	3200	[32]
brominated vinylated PIM-1 CO50	1.32	0.172	0.261		710	225	145	80	1600	[32]
Brominated vinylated PIM-1	1.58	0.163	0.238		675	305	130	90	1890	[32]
CANAL-Et	1	0.23	0.293	400	883	310	270	111	1340	[33]
CANAL-Et-iPr	0.97	0.26	0.306	530	1160	410	350	153	1830	[33]
CANAL-Me-iPr	0.97	0.27	0.315	730	1510	490	380	193	2210	[33]
MTTB	0.928	0.351	0.398		5897	864	196	144	3155	[34]
ITTB	0.93	0.349	0.397		5423	1012	276	187	3901	[34]
CTTB	0.931	0.349	0.396		5257	791	200	140	3087	[34]
PIM-M	1.021	0.283	0.337			897	254	230	4370	[35]
PIM-1	1.059	0.281	0.325			802	258	229	4259	[35]
PIM-BM-50	1.199	0.236	0.296			450	106	99	2369	[35]
PIM-BM-70	1.255	0.229	0.288			322	74	70	1689	[35]
PIM-BM-100	1.339	0.22	0.281			228	43	41	1110	[35]
CANAL-TB-2	0.945	0.291	0.358		3608	747	205	162	2520	[36]
CANAL-TB-1	0.983	0.286	0.346		2760	463	121	97	1678	[36]
PIM-1	1.06	0.26	0.324		3042	1717		629	5366	[37]
DNPI-50	1.11	0.25	0.308		1478	522		132	2627	[37]
TOTPI-50	1.19	0.22	0.270		2616	1198		413	4756	[37]
TOTPI-100	1.28	0.2	0.245		1368	642		190	3056	[37]
PIM-1	1.09	0.239	0.305		3949	1257	472	337	6957	[38]
PIM-PMDA-OH	1.18	0.2	0.256		190	30.5	7.7	6.9	198	[39]
PIM-6FDA-OH	1.22	0.23	0.277		259	45.2	9.1	10.8	263	[39]
PI-TB-2	1.1876	0.215	0.243	86	134	14	2.1	2.5	55	[40]
PI-TB-1	1.2592	0.223	0.255	376	607	119	27	31	457	[40]
TBDA2-SBI-PI	1.13	0.23	0.286	530	1155	240	65	49	1213	[41]
TBDA1-SBI-PI	1.15	0.21	0.273	398	915	190	45	35	895	[41]
KAUST-PI-1	1.09		0.299				97.5		2329	[42]
KAUST-PI-5	1.34		0.195				79.2		1560	[42]
PIM-1	1.063	0.26	0.322			1133		353	5366	[43]
TFMPSPIM4	1.089	0.26	0.321			737		217	3616	[43]
TFMPSPIM3	1.156	0.24	0.287			561		158	2841	[43]
TFMPSPIM2	1.196	0.22	0.270			308		75	1476	[43]
TFMPSPIM1	1.214	0.22	0.276			156		33	731	[43]
PIM-1	1.08		0.312		3408	1135	397	356	5135	[44]

PIM-1	1.06	0.26	0.324			1393	726	472	6538	[45]
TBPIM25	1.13	0.22	0.294			917	375	262	4441	[45]
TBPIM33	1.14	0.21	0.292			864	353	240	4353	[45]
PIM-EA-PI or PIM-PI-12	1.17		0.240	1580	4230	1380	457	369	7340	[46]
PIM-C1	1.065		0.330	3380	9870	3410	1310	980	18900	[47]
TPHI	1.339	0.17	0.214		27		0.09	0.19	4.7	[48]
TPHI-TR-350	1.318	0.181	0.208		61		0.41	0.46	16	[48]
TPHI-TR-400	1.287	0.196	0.227		520		8.3	16	320	[48]
TPHI-TR-450	1.273	0.204	0.235		810		4	8.4	270	[48]
TPHA	1.34	0.171	0.190		27		0.12	0.21	5.7	[48]
TPHA-TC-300	1.307	0.183	0.215		74		0.82	1.1	23	[48]
TPHA-TC-350	1.298	0.188	0.220		82		1.3	1.4	31	[48]
TPHA-TC-400	1.295	0.191	0.222		92		1.7	2.2	39	[48]
AOPIM-1	1.187		0.260		926	194	42.3	47.3	1073	[49]
TNTDA-DAT	1.092	0.214	0.278	776		159	24	32	728	[50]
TNTDA-MMBMA	1.109	0.196	0.265	526		96	16	21	438	[50]
TNTDA-TBDA1	1.12	0.181	0.255	258		75	12	15	397	[50]
TNTDA-FDBMA	1.128	0.203	0.263	584		122	22	27	569	[50]
PIM-1	1.066		0.320		3274	1396	789	483	9896	[51]
(PIM-PI)x-(6FDA-durene-PI)y (1:4)	1.29		0.238				69.4	74.1	1265	[52]
(PIM-PI)x-(6FDA-durene-PI)y (1:6)	1.3		0.237				68.8	73.7	1225	[52]
(PIM-PI)x-(6FDA-durene-PI)y (1:10)	1.32		0.229				60.2	64.4	1047	[52]
PIM-PI-1	1.16		0.260				37.3	41.1	495	[52]
PIM-EA(H2)-TB	1.06		0.308				62.6	53.1	1391	[53]
PIM-EA(H2)-TB	1.06		0.308	606	1630	350	77.6	62.8	1380	[53]
AZ-PIM-100	1.143		0.287				245		2894	[54]
AZ-PIM-R50	1.116		0.296				384		4246	[54]
AZ-PIM-A50	1.099		0.307				331		3879	[54]
Teflon AF 2400	1.744	0.33	0.278		2090	960	390	480	2200	[55]
Teflon AF 1600	1.836	0.31	0.239		550	270	80	110	520	[55]
Hyflon AD 60	1.93	0.23	0.204		210	67	12	24	150	[55]

4. Plots and tables

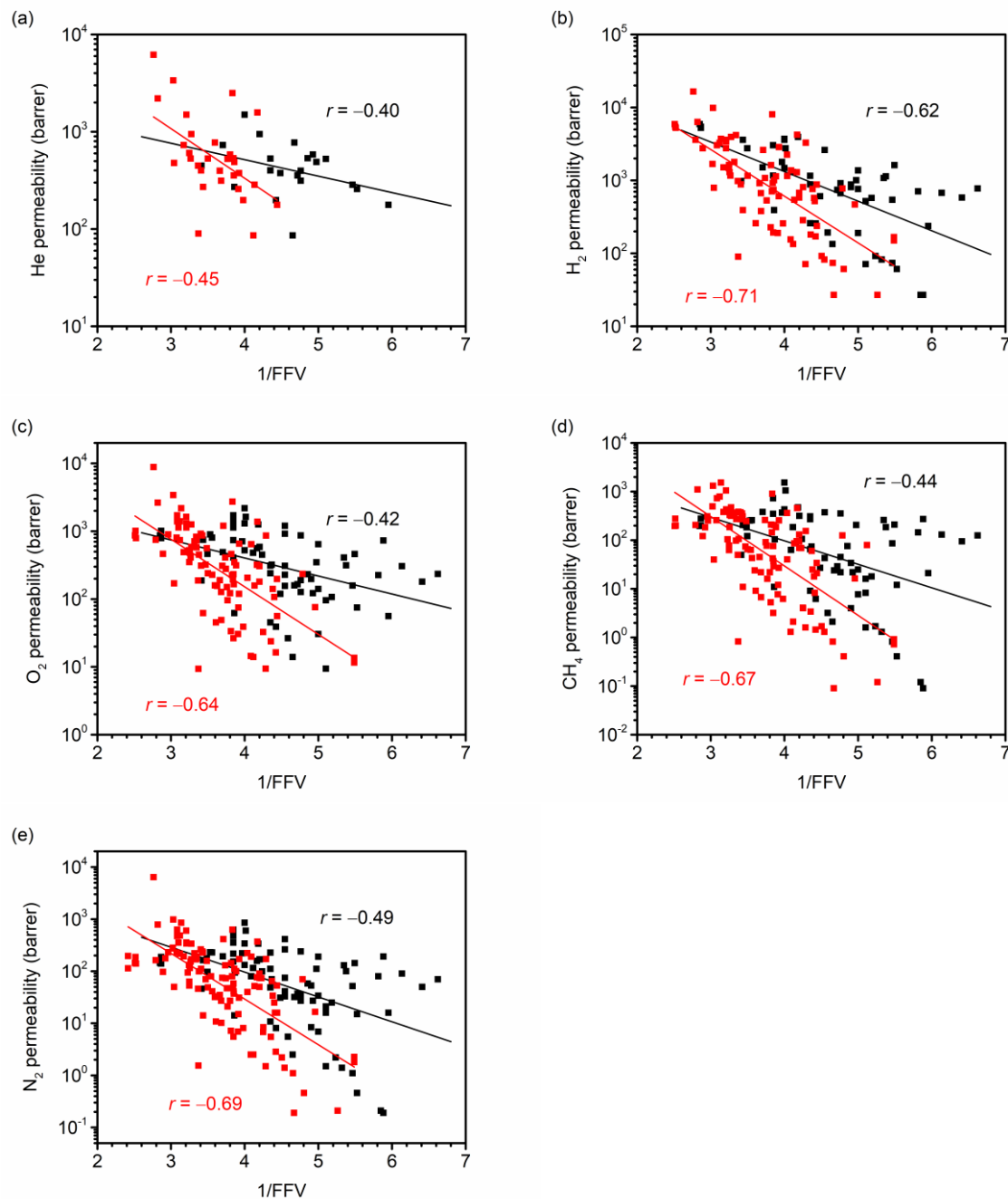


Figure S6: Plots showing overall correlations for $\ln(P)$ versus $1/FFV$ with their respective best-fit lines and Pearson correlation coefficients for (a) He, (b) H₂, (c) O₂, (d) CH₄, and (e) N₂. The red symbols represent FFV values calculated using the presented method and the black symbols represent reported FFV values.

Table S2: Table of the best-fit slopes and Pearson coefficients corresponding to Figure S5 and Figure 2.

Gas	Using new FFV (red symbols)		Using reported FFV (black symbols)	
	Slope	Pearson Coeff.	Slope	Pearson Coeff.
He	-1.268	-0.45	-0.389	-0.40
H ₂	-1.497	-0.71	-0.933	-0.62
O ₂	-1.645	-0.64	-0.610	-0.42
CH ₄	-2.369	-0.67	-1.112	-0.44
N ₂	-2.041	-0.69	-1.098	-0.49
CO ₂	-1.781	-0.67	-0.985	-0.48

5. MATLAB code for MC calculation of V_W using group P1 as an example

```
function bondi_mc
%atom identity
grp.a = ["C"
"C"
"C"
"C"
"C"
"H"
"H"
"H"
"H"
"H"
"H"
"H"
"H"
"H"
"H"
"H"];

%atom xyz coordinates in angstrom
grp.pos = [-4.344  -0.158  0.666
-3.207  0.631  0.004
-2.344  -0.321  -0.834
-2.34  1.285  1.087
-3.798  1.717  -0.903
-3.951  -0.949  1.326
-4.986  -0.642  -0.088
-4.984  0.499  1.277
-2.939  -0.809  -1.624
-1.903  -1.116  -0.21
-1.516  0.216  -1.324
-1.512  1.861  0.643
-1.899  0.529  1.758
-2.932  1.977  1.709
-4.425  2.419  -0.329
-4.428  1.278  -1.694
-3.005  2.303  -1.395];

%atoms excluded
grp.extra = sort([1
6
7
8
3
9
10
11]);

%removing excluded atoms from set
for i = length(grp.a):-1:1
    if ismember(i,grp.extra)
        grp.a(i) = [];
        grp.pos(i,:) = [];
```



```

end
end

%defining VDW radii and creating structure
vr = [1.1 1.7 1.55 1.5 1.47 2.1 1.8 1.75 1.85]; %H C N O F Si S Cl Br

vdr.H = vr(1);
vdr.C = vr(2);
vdr.N = vr(3);
vdr.O = vr(4);
vdr.F = vr(5);
vdr.Si = vr(6);
vdr.S = vr(7);
vdr.Cl = vr(8);
vdr.Br = vr(9);

%% Monte Carlo
n = 1000000; %number of points
k = 0; %counter

%find box dimensions and generate points
for i = 1:3
    box_dim(i,1) = min(grp.pos(:,i))-max(vr);
    box_dim(i,2) = max(grp.pos(:,i))+max(vr);
    pt_dim(:,i) = box_dim(i,1)+rand(1,n)*(box_dim(i,2)-box_dim(i,1));
end

%MC script; determine if distance between point and atom is less than the
%VDW radius of the atom
for i = 1:n
    pt = pt_dim(i,:);
    for j = 1:size(grp.pos,1)
        sep = dist(pt,grp.pos(j,:));
        if sep < vdr.(grp.a(j))
            k = k+1;
            break
        end
    end
end

%calculate Vw, variance in cm3/mol
box_vol = prod(box_dim(:,2)-box_dim(:,1));
MC_v_w = box_vol*k/n*1e-30*6.022e23*1000000;
MC_v_w_var = MC_v_w*(box_vol-MC_v_w)/n*1e-30*6.022e23*1000000;
end

function d = dist(a,b)
%a,b are xyz coordinates
d = sqrt((a(1)-b(1))^2+(a(2)-b(2))^2+(a(3)-b(3))^2);
end

```

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